

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations

Summer Semester July 2024

BT2003-1 – HEAT AND MASS TRANSFER (BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Linear Graph should be provided

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Unit of Thermal Conductivity in S.I system of units is _____
 A) $J/m^2 \cdot sec$ B) $J/sec. m. K$
 C) $J. sec/m. K$ D) $W/m^2 \cdot K$
- Heat transfer takes place according to which of the following law?
 A) Newton's second law of motion B) First law of thermodynamics
 C) Newton's law of cooling D) Second law of thermodynamics
- Which way is heat transfer believed to take place in a long, hollow cylinder that is kept at consistent but varied temperatures on its inner and outer surfaces?
 A) Radial only B) Unpredictable
 C) Axial only D) No heat transfer takes place
- The rate of heat transfer for a plane wall of homogenous material with constant thermal conductivity is given by which of the following equation?
 A) $Q = kA \Delta T / x$ B) $Q = 2k / \Delta T x$
 C) $Q = 2kA \Delta T x$ D) $Q = 2kAx / \Delta T$
- The energy transfer between the hot fluid and cold fluids is brought about by their complete physical mixing in
 A) Regenerators B) Direct contact heat exchanger
 C) Recuperators D) Boilers
- In how many categories heat exchangers are classified on the basis of direction of flow of fluids?
 A) 1 category B) 2 categories
 C) 3 categories D) 4 categories
- In a counter flow heat exchanger, a hot fluid is cooled from $100^\circ C$ to $70^\circ C$ by cold fluid and that gets heated from $25^\circ C$ to $55^\circ C$. The LMTD value of the heat exchanger is
 A) $45^\circ C$ B) $55^\circ C$
 C) $65^\circ C$ D) $75^\circ C$
- DPHE is used to cool fruit juice from the reactor to room temperature. The heat transfer co-efficient for DPHE is calculated by the equation $N_{Nu} = 0.023(N_{Re})^{0.8} (N_{Pr})^a$, what is the value of 'a'?
 A) 0.2 B) 0.4
 C) 0.1 D) 0.3
- For a binary mixture of species, A and B, the mass average velocity is defined as
 A) $\rho_a V_a + \rho_b V_b / \rho$ B) $2\rho_a V_a + \rho_b V_b / \rho$
 C) $\rho_a V_a + \rho_b V_b / 2\rho$ D) $\rho_a V_a + 2\rho_b V_b / \rho$

10. A solution contains 0.3 moles of solute A, 0.2 moles of B and 0.5 moles of C. What will be the mole fraction of A in the mixture?
 A) 1 B) 0.2
 C) 0.3 D) 0.5
11. The ratio of a mass concentration of species A to the total mass density of the mixture is known as
 A) Concentration B) Mass density
 C) Mole fraction D) Mass fraction
12. The units for diffusivity, D, are
 A) $m^2.s$ B) m^2/s
 C) s^2/m D) $m^2/L.s$
13. In Freundlich adsorption isotherm, the value of $1/n$ is always
 A) Between 2 to 4 in all cases B) 1 in the case for physical adsorption
 C) Between 0 to 1 in all cases D) 1 in the case for chemical adsorption
14. The equation represent the Freundlich adsorption isotherm is (Note: x/m is the mass of substance adsorbed per unit mass of adsorbent).
 A) $C^* = K (x/m)^n$ B) $C^* = K (x/m)^{1/n}$
 C) $x/m = K (C^*)^n$ D) $x/m = K (C^*)^{1/n}$
15. Stages in extractors are
 A) Equilibrium Stages B) Distribution Stages
 C) Theoretical Stages D) Equilibrium or theoretical stage
16. Find the sum of R & E if the sum of F & S is 80kmol (Where R: Raffinate, E: Extract, F: Feed, S: Solvent)
 A) 40kmol B) 20kmol
 C) 80kmol D) 60kmol
17. Which of the following is not separated through distillation process?
 A) Impurities in Sea water B) Milk and Water
 C) Acetone and water D) Aniline and chloroform
18. Find the slope of the operating line of the top section of Fractionator if Reflux ratio is 2. (Hint: use McCabe Thiele method is applicable)
 A) 0.77 B) 0.87
 C) 0.57 D) 0.67
19. The Enriching section operating line intercept at the origin in McCabe-Thiele method when
 A) Reflux ratio= 1.5 to 2 times Reflux minimum B) Reflux ratio= One
 C) Reflux ratio= Infinity D) Reflux ratio = Zero
20. Ratio of the vapour pressure of A to the vapour pressure of B at the same temperature is known as
 A) Relative Volatility B) Volatility
 C) Flash point D) Boiling point

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define (i) Kirchhoff's Law (ii) Stefan-Boltzmann Law (iii) Opaque materials	6	L*2	1	1
	b) Explain the procedure for determining the heat transfer area of a heat exchanger by LMTD approach and NTU approach.	10	L3	2	2
2.	a) A steam pipeline, 150/160 mm in diameter, is covered with a layer of insulating material of thickness 50mm. The temperature inside the pipeline is 393 K (120°C) and that of the outside surface of insulation is 313K (40°C). Calculate the rate of heat loss per 1 m length of pipeline. Data: k for pipe is 50 W/ (m. K) and k for insulating material is 0.08 W/ (m. K).	8	L3	1	2
	b) Explain the construction and working of plate type heat exchanger with a neat, labeled sketch. Mentions its application in bio-industries.	8	L2	2	3

3. a) A furnace is constructed with 225mm thick fire brick, 120mm of insulating brick and 225 mm of building brick. The inside temperature is 1200 K (927°C), and the outside temperature is 330 K (57°C). Find the heat loss per unit area and the temperature at the junction of the fire brick and insulating brick.

Data: k for fire brick = 1.4 W/ (m. K), k for insulating brick = 0.2 W/ (m. K), k for building brick = 0.7 W/ (m. K).

8 L3 1 2

- b) Compare forward feed arrangement with backward feed arrangement in case of multiple effect evaporation system.

8 L2 2 3

Unit – II

4. a) Define: (i) Mass fraction, (ii) Mole fraction, (iii) Molar concentration, (iv) Mass average velocity
- b) The equilibrium adsorption of acetone (M.Wt=58kg/kmol) vapour on an activated charcoal at 303K is given below

8 L2 3 1

g adsorbed/g charcoal	0	0.1	0.2	0.3	0.35
Partial pressure of acetone, mmHg	0	2	12	42	92

The vapour pressure of acetone at 303K is 283mmHg. A 1L flask contains air and acetone vapour at 1atm and 303K, with relative saturation of vapour of 35%. After 2g of activated charcoal has been introduced into the flask, the flask is sealed. Calculate the final vapour concentration at 303K and the final pressure neglecting the adsorption of air.

8 L3 4 3

5. a) Ammonia gas (A) diffuses through a non-diffusing nitrogen gas (B) under steady state conditions. The partial pressure of A at location 1 is 1.5×10^4 Pa and that at location 2 is 5×10^3 Pa (Pascal). Locations 1 and 2 are 0.15 m apart. The total pressure is 1.103×10^5 Pa and temperature is 298 K. Calculate the flux of diffusion of ammonia. Also calculate the flux of diffusion for equimolar counter diffusion considering that nitrogen is also diffusing. The diffusivity of ammonia at the prevailing conditions is $2.30 \times 10^{-5} \text{ m}^2/\text{s}$

8 L3 3 2

- b) List and explain the solvent selection criterion for extraction process.

8 L2 4 1

6. a) Explain the need for measuring $k_L a$ in fermenter. List different methods available for the measurement $k_L a$. Explain any one in detail.

8 L2 3 3

- b) For extraction of dioxane from water, benzene is used as an extraction solvent. At 298K the equilibrium distribution of dioxane between water and benzene is given below

Wt% dioxane in water	5.1	18.9	25.2
Wt% dioxane in benzene	5.2	22.5	32

1000kg of solution containing 25% dioxane is to be extracted with benzene to remove 95% of the dioxane. If extraction is done in 4 stage cross current unit using 750kg of pure benzene each time, determine the final raffinate concentration.

8 L3 4 3

Unit – III

7. a) Define: (i) Azeotrope (ii) Relative volatility (iii) Reflux ratio (iv) Volatility
- b) Explain in brief McCabe-Thiele method used for obtaining theoretical plates required for a given degree of separation.

8 L2 5 1

8 L2 5 3

8. a) With neat, labeled sketch explain the construction and working of Tray Dryer. Also list its advantage and disadvantages.

6 L2 5 3

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- b) A mixture of benzene and toluene containing 60mol % benzene is to be separated to give a product of 95mol % benzene and a bottom product containing 10mol % benzene. The feed enters a column at its bubble point. It is proposed to operate the column with reflux ratio of 2.5. It is required to find the number of theoretical plates needed and the position of the feed plate.

Equilibrium data:

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
y	0	0.21	0.375	0.5	0.6	0.7	0.77	0.83	0.9	0.95	1

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
